

DENVar: A Global Dynamic Estimate of NDVI Variance

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Abstract

Remote-sensing proxies of habitat productivity such as the Normalized Difference Vegetation Index (NDVI) have helped develop and support many hypotheses and projects. However, historically, users tended to focus solely on trends in mean productivity. In recent years, the variance around the mean (i.e., stochasticity) has gained more attention, particularly as climate change and human-induced rapid environmental change transform the ecosystems species have evolved in and adapted to. Still, calculating trends in NDVI stochasticity requires careful modeling, since stochasticity estimates are conditional on the estimated mean and the scale at which the data are collected and modeled. We thus present a new global Dynamic Estimate of NDVI Variance, *DENVar*. We estimate spatiotemporal trends in NDVI mean and variance using Generalized Additive Models, which provide a flexible yet transparent and interpretable model structure. We show that *DENVar* correlates nonlinearly with **XXX** and can be used to test many hypotheses related to environmental stochasticity, such as optimal foraging theory and regime shifts as well as for conservation. We provide a publicly and freely available repository for downloading rasters of daily and long-term averages of *DENVar* and the related estimates of mean NDVI.

41 useful refs:

- 42 • Global evaluation of four AVHRR–NDVI data sets: Intercomparison and
43 assessment against Landsat imagery: [https://www.sciencedirect.com/
44 science/article/abs/pii/S0034425711001957?fr=RR-2&ref=pdf_download&
45 rr=962beb22bcb58443](https://www.sciencedirect.com/science/article/abs/pii/S0034425711001957?fr=RR-2&ref=pdf_download&rr=962beb22bcb58443)
- 46 • Taylor’s law predicts unprecedented pulses of forest disturbance under global
47 change: <https://www.nature.com/articles/s41467-025-61585-5>
- 48 • AVHRR cloud detection using MODIS data:
- 49 • franch_30_2017
- 50 • villaescusa-nadal_modis-based_2022
- 51 • NDVI is available at a variety of scales [1]
- 52 • see `to-read` folder

53 Introduction

54 The Normalized Difference Vegetation Index [NDVI; 2,3] is a proxy of primary produc-
55 tivity used in a wide range of disciplines, including ecology [4,5], landscape phenology
56 [6,7], agriculture [8], forestry [9], and conservation [10,11]. High-resolution, remotely-
57 sensed NDVI data have allowed analysts to estimate plant biomass without the need
58 for fieldwork [12,13] and leverage such estimates for studying animal movement be-
59 havior [6,14]. Still, most work is limited to quantifying changes in and responses to
60 mean NDVI while ignoring changes in the variance around the mean.

61 Environmental stochasticity has been studied for decades [15–18], but recent work
62 has drawn more attention to the effects environmental stochasticity has at a wide range
63 of scales, from individual organisms [19,20] to across landscapes [11,21,22]. However,
64 estimating trends in NDVI variance requires more nuanced modeling than estimating
65 changes in mean NDVI, particularly when environments are temporally dynamic and
66 spatially heterogeneous. The scale at which processes are observed determine how
67 individuals’ perception of and ability to respond to change [23]. A forest fire is severely
68 disruptive and deadly for many insects [24], while birds and large mammals may not
69 struggle to escape the fire and or establish a new range [25,26]. Similarly, the effects
70 of the fire recovery will depend on its duration relative to other organisms’ phenology
71 and lifespan [27]. Therefore, distinguishing between a change in mean NDVI and
72 a stochastic fluctuation requires a conscious decision about the scale at which to
73 consider the data.

74 *Add a paragraph on what effects stochasticity may have/why it’s im-*
75 *portant?*

76 We present a new measure of environmental stochasticity, which we name the Dy-
77 namic Estimate of NDVI Variance (DENVar). We estimate DENVar using 44 years

78 of NDVI data from the AVHRR and VIIRS collection of sensors and Generalized Ad-
79 ditive Models (GAMs). Although satellites with AVHRR and VIIRS sensors provide
80 data at a coarser resolution (0.05°) than other satellites (e.g., 250m for MODIS), they
81 are the only sensors to provide data at a daily frequency and and for more than 40
82 years. Our use of GAMs provides a flexible yet interpretable approach for estimating
83 trends in mean and variance in NDVI without the rigidity of other more traditional
84 methods (e.g., ARIMA) or the opaqueness of machine learning models such as neu-
85 ral networks [28,29]. We illustrate the relationships between DENVar and multiple,
86 relevant ecological metrics, and we provide access to a free repository of daily, global
87 estimates of DENVar and the corresponding estimates of mean NDVI.

88 **Materials and Methods**

89 **Experimental and Technical Design**

- 90 • include a diagram to show the entire experimental design and illustrate the
91 most significant elements: materials, treatments, measurements, data collection,
92 methods of data analysis
- 93 • This will facilitate the editors and reviewers to understand and follow the whole
94 concept, design, and results.

95 **Choice of color palettes**

96 We represent NDVI using a modified version of Fabio Crameri’s divergent *bukavu*
97 palette [30,31], which has high-contrast for deuteranope and protanope vision. For
98 NDVI values between -0.1 and 1, the colors also have sufficient contrast for the colors
99 to also be distinguishable by strong tritanope and achromatic vision. Fig. A1 contains
100 an approximate representation of how each vision type would perceive each color

101 palettes used. We obtained all color palettes with three or more colors from the
102 `khroma` package [v. 1.14.0, 32] for R [v. 4.4.1, 33].

103 **Input data**

104 AVHRR data: from 1981-06-24 to 2013-12-31

105 VIIRS data: from 2014-01-01 to 2025-06-29

106 We downloaded NDVI image composites from satellites with AVHRR and VI-
107 IRS sensors [34,35] directly from the NOAA data servers (AVHRR data: [https://](https://ladsweb.modaps.eosdis.nasa.gov/archive/allData)
108 ladsweb.modaps.eosdis.nasa.gov/archive/allData; VIIRS data: [www.ncei.noaa.gov/](http://www.ncei.noaa.gov/data/)
109 [data/](http://www.ncei.noaa.gov/data/); Fig. A2)). We removed all cloudy pixels from AVHRR rasters and probably
110 or confidently cloudy pixels from VIIRS rasters. Additionally, we removed pixels with
111 excessively high NDVI values at high latitudes between September and March using
112 latitudinal cutoffs based on the monthly empirical cumulative distribution function
113 of NDVI across latitude (Fig. B2). We did not remove values at latitudes $< 60^\circ\text{S}$
114 since we exclude Antarctica from the analyses.

115 To ensure the number of rows for each dataset was below the maximum data frame
116 size in R ($3.17 \times 10^{10} > 2^{31} - 1 \approx 0.21 \times 10^{10}$), we reduced the dataset size by splitting
117 the dataset across four spatial groups (Fig. A4) and spatiotemporally averaged the
118 (cleaned) rasters by a factor of four. The final dataset thus had one raster every four
119 days with a spatial resolution of $0.05^\circ \times 4 = 0.2^\circ$. Despite the substantial reduction in
120 dataset size, the dataset creation and modeling were still challenging for the machine
121 used, which had 2.2 TB of RAM and an Intel Xeon Platinum 8462Y+ processor (32
122 cores).

123 The predictor data for the models included: integer day of year (1 to 366), integer
124 year (1981 to 2025), elevation above sea level (Fig. A6), and proportion pixel covered
125 by water (Fig. A7). We downloaded the global digital elevation model using the

126 `get_elev_raster()` function from the `elevatr` package for R [v. 0.99.0, 36] with a
 127 an original resolution of 0.076 degrees. To remove any biases towards negative NDVI
 128 values, we removed all raster cells with $> 40\%$ water cover before the spatiotemporal
 129 aggregation.

130 Modeling

131 We estimated spatiotemporal trends in mean NDVI using Generalized Additive
 132 Models (GAMs) via the `mgcv` package for R [v. 1.9-3, 29]. To reduce modeling fitting
 133 times with negligible losses to model accuracy, we used the `bam()` function with
 134 fast REstricted Marginal Likelihood (`method = fREML`) and covariate discretization
 135 [`discrete = TRUE`; 37,38]. Ideally, NDVI should be modeled using beta location-
 136 scale models (after the linear transformation $Y^* = \frac{Y+1}{2}$) to account for the fact that:
 137 (1) NDVI is bounded between -1 and 1, and (2) the variance in NDVI is dependent
 138 on the mean (and vice-versa), since ecosystems with very low NDVI (e.g., rock, ice,
 139 or concrete) or very high NDVI (e.g., dense forest) tend to have lower variance in
 140 NDVI. However, preliminary tests showed that fitting times for beta models were
 141 prohibitive, especially for beta location-scale models. In contrast, Gaussian models
 142 fit substantially faster and provided very similar spatialtemporal estimates of mean
 143 NDVI (see Appendix B). For each region show in Fig. A4, the model for the mean
 144 NDVI had the structure below:

$$\begin{cases} \text{NDVI} \sim \text{Normal}(\mu, \sigma^2) \\ \mu = \beta_0 + f_1(\text{prop_water}) + f_2(\text{elev_m}) + f_3(\text{doy}) + f_4(\text{year}) + f_5(\text{lat, long}) + \\ \quad f_{6T}(\text{doy, year}) + f_{7T}(\text{doy, lat, long}) + f_{8T}(\text{year, lat, long}) \end{cases}, \quad (1)$$

145 where, for a given (aggregated) pixel, `prop_water` is the proportion of water within

the pixel, `elev_m` is elevation above sea level in meters, `doy` is integer day of year, `year` is integer year, and `lat` and `long` are latitude and longitude, respectively. $f_j(\cdot)$ indicate the j^{th} marginal smooth functions, while $f_j T(\cdot)$ indicate the j^{th} tensor product interaction terms. Latitude and longitude were modeled using a spline-on-the-sphere basis [`bs = 'sos'; 29`], while seasonal terms used cyclical cubic splines (`bs = 'cc'`) with edge knots 0.5 and 366.5.

HERE

A second GAM estimated the variance in NDVI around the mean estimated by the model above. The model had an identical structure to the model for the mean, with the exception that (1) the response variable was the squared residual from the first model, such that the model estimated the mean squared residual (i.e., the variance) for a given point in time and space, and (2) a smooth term of the estimated mean NDVI. *Estimating the variance as the spatially or spatiotemporally varying mean squared residuals included the pixel-level squared bias.* The model is available below:

```
#' *CHANGE TO SOS BASIS*
m_var <- bam(
  e_2 ~
    biome +
    s(poly_id, bs = 'mrf', xt = list(nb = nbs)) +
    s(doy, biome, bs = 'fs', xt = list(bs = 'cc'), k = 10) +
    s(year, biome, bs = 'fs', xt = list(bs = 'cr'), k = 10) +
    ti(doy, year, biome, bs = c('cc', 'cr', 're'), k = c(5, 5)) +
    s(elevation_m, bs = 'cr', k = 5) +
    s(mu_hat, bs = 'cr', k = 5),
  family = gaussian(),
  data = d,
  method = 'fREML',
  knots = list(doy = c(0.5, 366.5)),
  drop.unused.levels = TRUE,
  discrete = TRUE,
  samfrac = 0.001,
  nthreads = future::availableCores(logical = FALSE) - 2,
  control = gam.control(trace = TRUE))
```

Results

figures:

- plot grid of mean over space, `s(year)`, `s(doy)`, `s(elev)`
- plot grid of DENVar over space, `s(year)`, `s(doy)`, `s(elev)`, `s(mean)`
- variance is likely overestimated in areas near sharp changes in NDVI, but this depends on scale
- histograms of DENVar around flat map of biomes (change colors)
- hex plots of DENVar with other variables (add temperature, annual precip, extreme weather events, fires, invasive species, max body size, disease occurrence, ...)

General results

- two Orthographic projections of DENVar: Northern and southern hemispheres on spring solstice

Correlations with other metrics

- temperature
- seasonal temperature range
- precip
- seasonal temperature range
- spp diversity
- gross primary productivity
- max animal weight
- <https://silvis.forest.wisc.edu/globalwui/>
- <https://silvis.forest.wisc.edu/data/dhis/>: NDVI16 (Normalized Difference Vegetation Index), EVI16 (Enhanced Vegetation Index), FPAR8 (Fraction absorbed Photosynthetically Active Radiation), LAI8 (Leaf Area Index), GPP8 (Gross Primary Productivity)

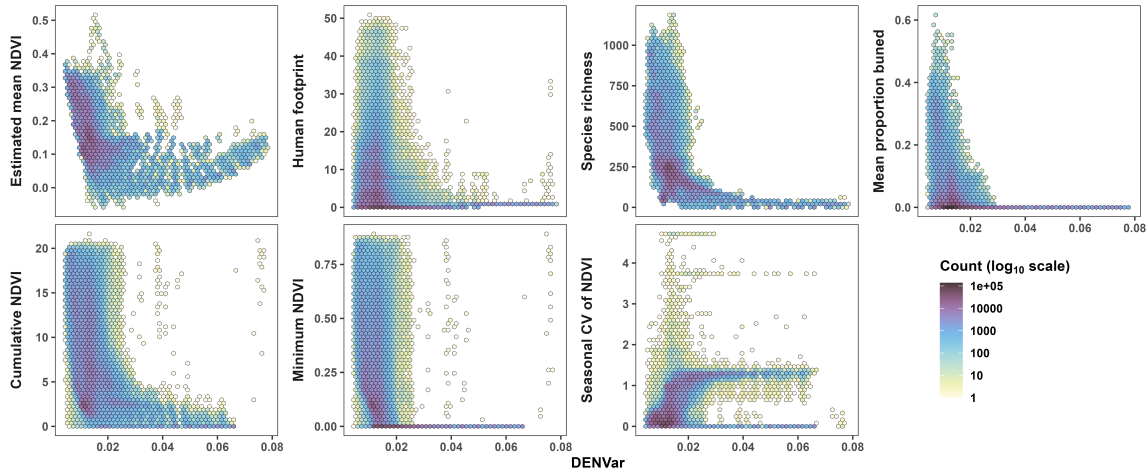


Figure 1: Hex plots of the relationship between the global Dynamic Estimate of NDVI Variance (DENVar) and common ecological metrics. The fill of the hexagonal cells represents the log-10-transformed number of points within the cell. Rowwise, the variables are: mean NDVI (as estimated in this manuscript); the machine-learning Human Footprint Index (ml-HFI) of Keys, Barnes & Carter (2021); the 2015 cumulative, minimum, and seasonal range in the Dynamic Habitat Index of XXX, (F).

How to obtain the data

Static rasters of mean and variance (1981-2025)

- for GIS users (tif)

Shiny app for obtaining values

- get values for specific coordinates: 1981-2025 average
- get values for specific coordinates and dates (upload as a CSV, download as a CSV)
- screenshot of the shiny app

194 Discussion

195 We have presented a new spatiotemporally dynamic metric of environmental
196 stochasticity, which we have called the Dynamic Estimate of NDVI Variance, DEN-
197 Var. In this section, we discuss the interpretation of DENVar in greater depth, and
198 we argue the advantages and limitations of the statistical methods used. We conclude
199 by suggesting some use cases of DENVar within animal movement ecology, landscape
200 ecology, and phenology, but we invite readers to consider how DENVar may be of use
201 in their field and adjacent disciplines.

202 Why use complicated statistics instead of machine learning?

203 Admittedly, the title of this section is misleading for two reasons. Firstly, some
204 may argue that Generalized Additive Models are a form of machine-learning (here
205 abbreviated as ML), but we intend to contrast GAMs with more “black-box” types of
206 machine learning, such as neural networks and random forests. The second reason for
207 which the title is misleading is that it implies that machine learning is not complicated,
208 or at least that it is less complicated than the models we used. However, we argue
209 that while traditional statistical models may seem more intimidating to some, their
210 complexity is what allows analysts to produce interpretable results and measures of
211 uncertainty [29]. And while the theoretical basis for GAMs can be discouraging for those
212 who do not have a strong background in mathematics or statistics, outputs from GAMs
213 can be presented in an accessible manner even to those without a strong quantitative
214 background through the use of well-designed figures, especially in a Bayesian context
215 (when the aim is to estimate a parameter). In contrast, while ML methods will often
216 give understandable results, it may not be possible to discern *why* such results were
217 “chosen” by the model. Instead, GAMs produce flexible yet interpretable estimated

218 effects that can be represented with relatively simple figures, even if the mathematical
219 formula(e) for the model may be complicated.

220 Ideally, a model should estimate or explore the relationships among variables
221 such that results can be interpreted in a transparent and unambiguous manner, such
222 that readers can scrutinize the underlying assumptions, methods, and decisions. We
223 recognize that the assessing GAMs requires a certain level of expertise, but we hope
224 the readers can appreciate the option to assess the GAMs more easily than a series
225 of consecutive, weighted regressions (as in the case of neural networks). In contrast,
226 GAMs learn from data with no initial starting assumptions other than user-imposed
227 limitations of wiggleness, which are clearly outlined in each basis size of the models'
228 terms. The structure and wuggleness of GAMS can thus be grossly designed based on
229 the system's complexity and structure while allowing the data to inform the model
230 trends, wiggleness, and coefficient size.

231 **Interpreting DENVar: the importance of scale**

- 232 • smoothness of mean impacts smoothness and size of variance: depends on scale
233 of interest (e.g., animals' ability to respond to, learn, and predict changes in
234 the mean). Changes not attributed to trends in the mean are stochastic
- 235 • DENVar is a good metric as long as the spatiotemporal scale of interest is similar
236 or larger to that of DENVar. May not work well for small-scale processes such
237 as the movement dynamics of a small rodent, the phenology of a small orchard
238 when spatiotemporal heterogeneity is finer than DENVar's resolution

239 **Environmental stochasticity and animal movement ecology**

240 The field of animal movement ecology has long shown that animals alter their
241 movement in response to resource abundance or availability [39–42]. More recently,

242 movement ecologists have started focusing more on resource stochasticity [e.g., 16,20],
 243 but it is worth distinguishing between resource stochasticity and seasonality [43].
 244 Seasonality is generally best interpreted as a cyclical change in the mean that many
 245 animals can learn to predict, as in the case of green-wave surfing [44,45]. In contrast,
 246 environmental stochasticity refers to unpredictable changes, such as forest fires [ref?],
 247 sudden snow storms [46], and unusual phenology [ref?]. Animals also respond to
 248 stochastic changes [19], but their ability to adapt depends on behavioral plasticity
 249 [47,48], and memory [49–53].

250 [54] tipping points

251 [55]

252 extreme events, black swans [56,57]

- 253 • migration routes are key habitat (Ortega et al., in prep), especially in stochastic
 254 environments
- 255 • can fail to adapt to changes in conditions [58]
- 256 • <https://www.sciencedirect.com/science/article/pii/S0960982220308484>
- 257 • <https://onlinelibrary.wiley.com/doi/full/10.1111/gcb.15169>
- 258 • industrial development affects animals' ability to respond to both predictable
 259 and unpredictable change ([https://www.nature.com/articles/s41559-022-](https://www.nature.com/articles/s41559-022-01887-9)
 260 [01887-9](https://www.nature.com/articles/s41559-022-01887-9))
- 261 • effects of sudden deep snow in red desert on proghorn ([https://www.](https://www.sciencedirect.com/science/article/abs/pii/S0960982225002957)
 262 [sciencedirect.com/science/article/abs/pii/S0960982225002957](https://www.sciencedirect.com/science/article/abs/pii/S0960982225002957))
- 263 • <https://www.nature.com/articles/s41467-023-37750-z>
- 264 • <https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecy.4238>
- 265 • detecting regime shifts [in lakes: 59]

266 **Environmental stochasticity and population dynamics**

267 read [17]

268 Environmental stochasticity reduces habitats' energetic balance and carrying capacity
269 [60] as individuals struggle to rely on unpredictable resources and increase the insta-
270 bility of populations and communities [17]. Additionally, stochastic environments
271 reduce the ability of specialist species to adapt and evolve while selecting for more
272 generalist species [15]. Consequently, stochastic environments tend to have lower
273 species diversity [11].

274 **Environmental stochasticity and fire ecology**

275 Recent work by [devries?] has suggested that NDVI stochasticity is an important
276 metric for evaluating forest recovery after large fires, while [collinson?] has suggested
277 that variance in NDVI is an important predictor of the occurrence of forest fires due
278 to ignition by lightning.

279 **Environmental stochasticity and phenology**

- 280 • [61]
- 281 • effects can be compounded by cumulative effects (e.g., loss of species diversity,
282 see [62])
- 283 • climate change affects plant phenology, but changes in situ may not match
284 experimental estimates [63]
- 285 • reproduction of painted lady butterfly [64]
- 286 • [65]

Useful references

- [66]: in the dry season preceding the rut topi antelope (*Damaliscus lunatus*) respond to forage availability at the large scale but not the fine scale

- [67]

- [60]

- [48]

- [68]

- [4]

- [69]

- [43]

- [6]

- [70]

- [71]

- [72]

- [73]

- [74]

- [75]

- calibrating across sensors: <https://ieeexplore.ieee.org/abstract/document/6809983>

TO READ: * Site fidelity as a maladaptive behavior in the Anthropocene: <https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/fee.2456> * Pettorelli N., Vik J.O., Mysterud A., Gaillard J.-M., Tucker C.J. & Stenseth N.Chr. (2005). Using the satellite-derived NDVI to assess ecological responses to environmental change. Trends in Ecology & Evolution 20, 503–510. <https://doi.org/10.1016/j.tree.2005.05.011>

311 supplementary material:

312 • hex plots of pred vs obs for mean and var

313 • histograms of mean around flat map of biomes (change colors)

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